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Original Manuscript

Are faculty members capable of finding the needed information? A study of selected universities of Jammu and Kashmir

Information Development
1–15
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DOI: 10.1177/02666669241286056
journals.sagepub.com/home/idv
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Abstract

We all need information and there is so much of it available in both physical and digital media. However, to find the needed information, capability of searching; knowledge of retrieval tools like catalogue; knowledge of Boolean operators, wildcards and truncations; and knowledge to different sources of information like bibliography, encyclopaedia, journals, etc. are equally important. The current study set out to investigate the faculty members' searching capabilities in four of Jammu & Kashmir's top institutions. The study used questionnaire as a tool to gather data from randomly identified population. After applying appropriate statistical techniques, the study revealed that the faculty members are not able to cross the half mark of correct answering for some basic questions posed to them. Eventually, the study suggests to conduct user awareness programmes since these programmes are significantly associated with the correct responses given by the faculty members in the universities studied.

Keywords

information literacy, computer literacy, digital literacy, library orientation, information-seeking behaviour, search and retrieval, user awareness

Submitted: 17 April 2024; accepted: 4 September 2024

Introduction

We are surrounded by information, which is crucial to our culture and way of life. Fundamentally, people are knowledge seekers and users. Humans have always required information to guide their actions and decisions. The volume of information and the complexity of information systems have increased due to advancements in information technology. Since Information Communication Technology (ICT) has made clear impact on various aspects of life, libraries are no different to it (Krubu and Osawaru, 2010). A growing number of information resources are now being converted into digital files, and libraries are concentrating on digital advancement as they use the newest technologies to enhance and automate their library operations. They construct databases and provide users with access to web-based data hosted on servers located throughout the globe. Libraries nowadays aren't limited to creating databases and facilitating access to information stored in remote locations. Rather, they are concerned with providing digital services consistently with user satisfaction. Hence, users need to be taught how to search and analyse

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information, how to use catalogues online and electronic resources, etc. and ultimately how to evaluate the sources of information. To effectively search for, locate, evaluate, apply information, as well as to conduct independent study, and lifelong learning, one must possess information literacy skills (Hemavathi, 2019).

Information literacy (IL) is a collection of abilities and knowledge that enables us to filter out unnecessary information in addition to locating, assessing, and using the information we need (Eisenberg, 2010). To be more specific, information literacy is the capability to ascertain the quantity of information required, discover and assess information, and decide how to utilise the information in a morally and legally responsible manner (Ramesha, 2008).

Chartered Institute of Library and Information Professionals (CILIP) (2018) defined information literacy as:

> Information literacy is the ability to think critically and make balanced judgements about any information we find and use. It empowers us as citizens to reach and express informed views and to engage fully with society.

In the current era of ICT, users must be taught how to use information effectively starting in preschool and continuing through college. Librarians play an important role in IL (Pullman, 2016) because they teach people how to use information effectively and efficiently at all levels of schooling so that they can utilise the information in an effective manner. Experts, educators, and students must all learn how to utilise information critically in a world where there seems to be an endless supply of information. Learning the abilities required to locate information from a variety of sources is just as important as knowing how to use the Internet to obtain information. Whatever information is discovered is irrelevant; what matters is how it is applied to complete the task or research. This is why information literacy skills are most important in the current era of ICT.

Faculty members must increase their knowledge in the modern world so that they can disseminate information effectively. They must be able to determine what information they require and in what quantity. They should be able to locate information sources that are both print and non-print, and learn how to use the appropriate resources and techniques to find them. They should be able to locate the information they require, download it, and save it to the preferred storage medium. They should be able to evaluate the information so as to assess the material according to criteria such as relevancy, accuracy, timeliness, authenticity, and authority. To create new knowledge or alter what they already know, academics should be able to integrate the new information with what they already know. They should maintain academic integrity by abiding by professional ethics, being mindful of copyright and plagiarism. They will undoubtedly play more essential, visible, and transparent roles in academic institutions if they possess all of the aforementioned abilities (Singh, 2008).

In this backdrop, since there has not been any study on information literacy of faculty members in Jammu & Kashmir, it is essential to know how they search for information. Furthermore, it is also important to research whether the faculty members know about the basic technical things of a library so that they can use the library resources comprehensively. However, it is imperative to find relationship between the user awareness programmes attended by the faculty members and capability to answer correctly so that it can be ascertained whether there is need of conducting such programmes or not.

Review of related literature

Numerous studies have been conducted on information literacy, computer literacy, Internet literacy, information-seeking behaviour, etc. In order to retrieve the relevant literature related to information literacy, the investigators thoroughly searched the literature and looked through databases like Google Scholar, Shodhganga, Electronic Management Research Library Database (EMERALD), Science Direct, JSTOR, Shodhganga, and the Internet. The literature was retrieved after searching different databases with terms like, "information literacy", "information seeking behaviour", "use of electronic resources", "information seeking in web environment", "information searching", "digital information seeking behaviour", etc. Few bibliometric studies provide an overview of the literature published on information literacy. One such study has been carried out by Ali et al. (2022) on Web of Science (WoS) indexed papers. The authors discovered a total of 4079 publications and among those 2168 publications were on "information literacy and libraries". Furthermore, 3662 publications were found to be in English with University of Illinois being the highest contributing institute and *Journal of Academic Librarianship* as the top journal.

The literature retrieved after searching the databases was broadly categorised under the following headings:

ICT literacy skills

In order to become information literate, each individual needs to have some ICT skills. ICT has remained the focus of many studies in the past. Okello-Obura and Ikoja-Odongo (2010) studied Library and Information Science (LIS) postgraduate students at Makerere University in Uganda. The study showed that most LIS postgraduate students possess different ICT literacy skills. However, Prakash and Kannappanavar (2016) evaluated the ICT literacy levels of faculty members and researchers in the fields of Science and Technology in the Universities of Karnataka State. It was found that the use of ICT tools and applications was satisfactory. Both research scholars and faculty members were proficient in retrieving desired results. Similarly, Dr Babasaheb Ambedkar Marathwada University Library, Aurangabad was the subject of a study by Padme and Dhande (2014). According to the study, 8.1% of the participants were unable to use a computer. The research showed that 5.1% of all respondents lacked the necessary skills to use the Internet. So, most of them had good ICT skills.

Using Boolean operators, truncation, etc.

Knowledge of search strategies is must for refining the results and to retrieve the pinpointed information. Liang and Tsai (2009) looked at how medical students use online medical information. The group was made up of 534 medical students from 6 medical school in Taiwan. From the study, it was found that students of different genders used different search strategies. Nisha and Varghese (2015) looked into searching capability of faculty members in Kerala. This study revealed that faculty members do not know various search techniques like Boolean operators, proximity operators, truncation, etc. Furthermore, the study revealed that 48% of faculty members know the basics of advanced searching. In contrary, the study of Onuoha and Molokwu (2016) revealed that majority of the students can easily conduct searches using Boolean operators in Colleges of Agriculture, South-West, Nigeria. Likewise, Bhat and Ganaie (2016) conducted a study on the users of the Dr. Y.S. Parmar University of Horticulture and Forestry (DYSPUH & F). The study reveals that less than 50% of users can utilise Boolean operators while as majority of users are not proficient in the sophisticated search techniques.

Khan and Khanam (2014) looked into how Aligarh Muslim University's (AMU) faculty members who teach Social Science and the Arts used the Internet's resources and services. The study reveals that most of the faculty members from Social Sciences use keywords (39%) for searching followed by phrase searching (23%) and Boolean logic (21%). However, the faculty members from Arts mostly use keywords (41%) followed by Boolean logic (31%) and phrase searching (15%). Likewise, researchers Majid et al. (2011) deduced from their study that respondents were using quick/basic search option more frequently than the advanced search. The AND operator was used the most and the NOT operator the least. These operators are not usually used by respondents when conducting literature searches. Other search options like truncation/wildcards, and proximity operators weren't used very often. Lata and Sharma (2013) tried to find out how well the teachers and students are information literate at Postgraduate Institute of Medical Education and Research, Chandigarh and Pt. B.D. Sharma University of Health Science, Rohtak. The study discloses that only a small number of respondents from both medical schools knew about the Boolean operator OR. Bhat et al. (2015) revealed that most of the users (50%) have not used the Boolean operators for searching while as 87.2% have never used wild cards, 86.7% have never used truncation and 74% have never used phrase searching.

Likewise, information literacy was evaluated by Munshi and Nagar (2016) among the postgraduate students of AMU to understand their fundamental requirements and information search techniques. The majority of students access the library material through Online Public Access Catalogue (OPAC). However, Griffiths and Brophy (2005) found only 10% users using the university library catalogue. In addition to this, the keyword search is a common technique for finding information online (Munshi and Nagar, 2016). Likewise, Bhat and Ganaie (2016) also found majority of users starting their information searches with the title (79.8%) followed by keywords (58.7%) at Dr Y.S. Parmar University of Horticulture and Forestry (DYSPUH & F).

Using search engines

Nazim (2008) looked at how Internet users at AMU search for the information. The study reveals that

respondents mostly used Internet search engines to find the information. Pradhan (2017) conducted study on search strategy of dental students of Institute of Dental Sciences, Siksha 'O' Anusandhan (SOA) University. It was observed from the study that most of the students were using the multilingual search engine. According to Nachiappan and Jeyshankar's (2015) study, web search tools are used by students and scholars of Alagappa University. The survey showed that search engines were used on the web often by the scholars as well as by the students. Nearly 90% of the respondents start their browsing through search engines. Similarly, according to Dee and Stanley's (2005) research about one-third of respondents, regularly utilise search engines on the web.

Ravichandran et al. (2013) looked at information literacy among the students in Chennai. The study was done with Engineering, Arts and Science students in Chennai among whom were undergraduates as well as postgraduates. Respondents from the Arts and Sciences who filled out the survey used Google more to find information on the Internet. Nagaraju and Roja (2014) polled at Atreya Ayurvedic Medical College & Research Centre, Doddaballapur, Karnataka, to find out how well-informed the students are about web searching. The study reveals that Google was being used by 86% of the respondents. Likewise, a study on use of search engines was done by Munshi and Nagar (2016) among the postgraduate students of AMU. The study found Google as the most popular search engine. Bhat et al. (2015) also revealed that most of the users (98.6%) use Google followed by Yahoo (39.4%). Similarly, Griffiths and Brophy (2005) focused on how students search the information. From the study, it was found that 45% of students search for information on Google. In order to determine the most popular searching methods, and strategies used by users of the Dr Y.S. Parmar University of Horticulture and Forestry (DYSPUH & F), Bhat and Ganaie (2016) conducted a study. Majority of the library users prefer to utilise search engines (63.5%) and Google (100%) consistently ranks as the best search engine followed by Yahoo (26%). Similarly, Salehi et al. (2018) also carried out a study on use of search engines. It has been seen that students rely more and more on web searches for learning. Most of the students who took part in the survey said that they use Google as their main or only way to find information. Bhatia and Rao (2011) studied the information-seeking behaviour of students at DevSamaj College in Chandigarh. Randomly, 100 students who visited the library were given a questionnaire, and 64.0% of them filled it out. Most users who use e-resources do so through search engines to get the information they need. In their study, Kumar and Bhandi (2015) discovered that 67.9% of postgraduates and 71.4% of faculty members favoured using web search engines to find e-resources.

Information literacy programmes

Studies have revealed that there is a clear connection between use of information and knowledge of ICT. One such study of Majid and Abazova (1999) looked into the link between how well teaching staff at the International Islamic University in Malaysia knew how to use computers and how much they used information sources through Internet. The study showed that there is a statistically significant link between knowing how to use computers and using Internet information sites and services effectively. Academics who study computers use Internet sources of information more often. Hence, the more informed the users are, the more they can use different resources. In order to make users more informed, we need to focus on IL programmes.

Regarding the library orientations programmes, researchers have delved to gauge the presence of users in these programmes. Dsilva (2017) found around 86.5% of respondents of their study participated in such programmes similar to the findings of Jayapal (2016) who found 83.8% users who had attended such programmes. However, the study of Lata (2014) reveals a lower percentage of only 60.2% of the faculty members in the medical colleges of Haryana, Punjab and Chandigarh who have attended IL programmes. Similarly, Sarman (2013) reveals only 44.6% respondents who have attended workshops and seminars. Likewise, Logeswari et al. (2021) found even lower percentage of 35.3% users who have attended such workshops and seminars.

Based on the studies carried out by different researchers, they have suggested to conduct IL programmes for better competency among the users since they don't use much of the sources on the web (Khan, 2014). Bansode and Pujar (2008) suggest that the researchers need to be trained on how to use the Internet so that they could use the right tools and methods to find information. According to Batabyal and Rath (2017), the academic and university libraries should carry out seminars, workshops, and training programmes by assembling a team of specialists for

offering such programmes connected to information literacy. Likewise, Parameshwar and Patil (2009) suggested that orientation and training programmes should be conducted on how to use Internet tools and services of library.

As a result, while users possess ICT abilities, the analysis of related literature demonstrates that they frequently lack knowledge of using Boolean operators, truncation, phrase searching, etc. To find information, the majority of the users, however, choose to utilise search engines like Google which retrieve results in millions with low precision. As a result, conducting information literacy programmes has become essential in order to teach users how to search the literature in various ways, assess information sources, and use information efficiently.

Objectives of the study

The objectives of the study are:

1. To determine the faculty members' aptitude for finding and retrieving the information precisely.
2. To reveal the know-how of faculty members regarding library catalogue, bibliography, thesaurus and call number.
3. To determine the faculty members' level of understanding regarding plagiarism.
4. To reveal the relationship between user awareness programmes and capability of answering the basic library-related questions correctly.

Methodology

Quantitative methodology was used for the study in which survey method was used to get the information from faculty members. In this study, a well-planned and organised questionnaire was used to gather information. The research questions were made based on the goals of the study.

Scope

There are many higher education institutions in J&K which include universities, colleges and research centres as well. However, in order to confine the scope of the study, it was decided to limit the study to general universities in Kashmir Division of the J&K UT only so as to make the study manageable. The four universities, which are almost all general in nature and not specialised towards any particular subject, considered for the study include:

(A) University of Kashmir (UoK)
(B) Islamic University of Science and Technology (IUST)
(C) Cluster University Srinagar (CUS)
(D) Central University of Kashmir (CUK)

Furthermore, the selection was made because of the reason that IUST and CUK are not far from Srinagar and UoK and CUS are in Srinagar. So, it was convenient for the investigators to often visit these universities and collect the filled questionnaires from the faculty members which is always a tedious job.

Data collection tool

Although many data collection tools are used to meet the requirements of a study. These tools include: interview, questionnaire, schedule etc. After assessing the merits and demerits of various tools and based on the studies carried out so far, it was found that questionnaire is most suitable for collecting data from the faculty members since it gives time to the respondents to answer and is free from bias of the investigator (Kothari, 2004). Questionnaire has also been used by Baro and Fyneman (2009); Rehman and Alfaresi (2009); Baro et al. (2011); Opesade and Tiamiyu (2022) in their survey-based studies.

The faculty members of four selected universities were given the closed-ended questionnaires after finalising the same with the help of experts from Library and Information Science background so as to remove any errors, mistakes and wrongly framed questions. The questionnaire consisted of few personal questions, ten multiple-choice questions with one correct answer and one dichotomous question (Yes/No) as well.

Sample of the study

Sampling may be described as choosing a portion of an aggregate or totality to base a conclusion or judgement on the aggregate or totality. In other words, it is the process of learning about a population as a whole by looking at a subset of it. A researcher quite often selects only a few items from the universe for study purpose (Kothari, 2004).

All the permanent faculty members in the four universities under study are designated as Professors, Associate Professors and Assistant Professors while few other are appointed on contractual basis. The investigator sent the questionnaire to all the teachers irrespective of designation and qualification. A list of all the faculty members was prepared after visiting

Table 1. University-wise distribution of responses

Name of the university	Questionnaires distributed	Questionnaires received	Percentage of response
University of Kashmir (UoK)	150	113	75.33
Islamic University of Science & Technology (IUST)	100	66	66.00
Cluster University Srinagar (CUS)	150	85	56.67
Central University of Kashmir (CUK)	80	54	67.50
Total	480	318	66.25

the websites of all the universities under study. The total number of faculty members came out to be 1288 as on January 2024. Since the population (1288) is large, the investigators decided to choose a subset by randomly selecting faculty members from the above-mentioned universities by going for sample size calculation.

Sample size calculation

Creative Research Systems' online sample size calculator (https://www.surveysystem.com/sscalc.htm) was used to calculate the sample size of the population. Since there are 1288 faculty members as calculated above, the online calculator found that the minimum sample size for the faculty members was 296. The details are given below:

✓ Confidence Level = 95%
✓ Confidence Level = 5
✓ Number of People = 1288
✓ Need a sample size of 296

Although the sample size calculated was 296, the investigators managed to retrieve back 318 valid questionnaires and used all these questionnaires for the study due to the reasons quoted by Andrade (2020):

> A sample that is larger than necessary will be better representative of the population and will hence provide more accurate results.

The data from 318 questionnaires was used in the analysis.

Data collection

The questionnaires were distributed from January 2023 to October 2023. A total of 480 questionnaires were distributed among the faculty members of four universities undertaken for the study but only 318 valid questionnaires were received back generating a total response rate of 66.3%. Highest response was observed in University of Kashmir (UoK) (75.3%) followed by Central University of Kashmir (CUK) (67.5%), Islamic University of Science & Technology (IUST) (66%) and Cluster University Srinagar (CUS) (56.7%). The response received is tabulated in Table 1.

Data analysis

The collected data was analysed in a number of ways. It was coded and put into a Statistical Package for Social Sciences (SPSS) (version 21.0), and Microsoft Excel was used to make basic analysis. Microsoft Excel helped in calculation of percentage and total in most of the tables while as SPSS helped in understanding the relationship with chi-square test. The decision to use Microsoft Excel was taken after consulting the studies of Nachiappan (2015); Kumar (2018) while as SPSS has been used by Hemavathi (2019); Nachiappan (2015); Lata (2014); Dsilva (2017); Shivkumara (2021) and Kumar (2018) to derive relationships between different variables.

Furthermore, the Chi-square test and Crosstabs (Cramer's V) were used in SPSS to reveal the relationship between variables. The decision to use the Chi-square test is taken after consulting the studies of Kamatchi (2013); Asha (2019); Kumar (2018); Hemavathi (2019); Nachiappan (2015); Lata (2014); Shanmugam (2005) and Shivkumara (2021).

Retrieving basic information

All the faculty members were asked few questions so that the literacy level of faculty members is assessed on the basis of correct responses given by them.

The overall figure reveals that only 124 (39.4%) faculty members responded correctly to the question posed to reveal their knowledge regarding encyclopaedias. The figures indicate that most of them are unaware of encyclopaedias which provide basic information related to any topic. However, Batabyal (2018)

found 64% respondents aware of finding background information from encyclopaedia while as Lata (2014) found 49.8% faculty members aware of encyclopaedias providing basic information in Haryana, Punjab and Chandigarh and Rehman and Alfaresi (2009) found only 33.7% aware of encyclopaedias in Kuwait.

University-wise response to the question revealed that most of the faculty members of UoK have correctly answered the question when compared with other universities followed by CUK, CUS and IUST, respectively (Table 2).

It is, thus, inferred that faculty members are not aware of retrieving basic information from encyclopaedias.

Retrieving information related to research

The study reveals that around 65.8% of the faculty members correctly answered to the question posed related to retrieving most current information. The results reveal that most of them are aware about journals providing current information. However, Sarman (2013) found that only 56.3% of the respondents answered correctly to the question related to the source used to find most recent information while as Lata (2014) found around 77.3% faculty members in the medical colleges of Haryana, Punjab and Chandigarh searching for current information in the journals.

University-wise response to the question revealed that among the four universities, UoK (70.8%) faculty members are much aware about the journals providing most current information on a topic followed by CUS, IUST and CUK (Table 3).

It is inferred that most of the faculty members are aware about journals but the faculty members of UoK are more aware about journals providing latest information than other three universities.

Locating a book on the shelves of a library

The study reveals that majority of the faculty members are unaware of call number and try to locate the book on the shelves by title which is often a tedious job when classification scheme is in place in all the universities selected for the study. However, Batabyal (2018) found 37.4% respondents aware of call number while as Lata (2014) revealed identical figures in which only 28.9% faculty members were found to be aware of call number.

University-wise response to the question revealed that more than half of the CUK (53.7%) faculty members are aware about call number used to locate a book on the shelves of a library while as the faculty members of other universities are mostly unaware of this thing (Table 4).

It is clearly inferred from the results that faculty members are not aware about call number. However, about half of the faculty members of CUK are aware of call number.

Table 2. University-wise retrieving basic information

University	Library catalogue	Bibliography	Journal	Encyclopaedia	Total
UoK	13 (11.7)	7 (6.3)	45 (40.5)	46 (41.4)	111
IUST	7 (10.6)	8 (12.1)	28 (42.4)	23 (34.8)	66
CUS	8 (9.5)	3 (3.6)	40 (47.6)	33 (39.3)	84
CUK	3 (5.6)	7 (13)	22 (40.7)	22 (40.7)	54
Total	31 (9.8)	25 (7.9)	135 (42.9)	124 (39.4)	315

Values in the parentheses indicate percentage.

Table 3. University-wise retrieving information related to research

University	Encyclopaedia	Bibliography	Journal	Book	Total
UoK	26 (23)	6 (5.3)	80 (70.8)	1 (.9)	113
IUST	12 (18.2)	8 (12.1)	43 (65.2)	3 (4.5)	66
CUS	19 (22.9)	3 (3.6)	55 (66.3)	6 (7.2)	83
CUK	13 (24.1)	10 (18.5)	30 (55.6)	1 (1.9)	54
Total	70 (22.2)	27 (8.5)	208 (65.8)	11 (3.5)	316

Values in the parentheses indicate percentage.

Table 4. University-wise locating a book on the shelves of a library

University	Title	Author	ISBN	Call number	Total
UoK	75 (66.4)	9 (8)	9 (8)	20 (17.7)	113
IUST	45 (68.2)	8 (12.1)	6 (9.1)	7 (10.6)	66
CUS	62 (73.8)	9 (10.7)	5 (6)	8 (9.5)	84
CUK	15 (27.8)	9 (16.7)	1 (1.9)	29 (53.7)	54
Total	197 (62.1)	35 (11)	21 (6.6)	64 (20.2)	317

Values in the parentheses indicate percentage.

Table 5. University-wise knowledge of library catalogue

University	Periodical index	Internet	Library catalogue	Bibliography	Total
UoK	11 (10.1)	39 (35.8)	54 (49.5)	5 (4.6)	109
IUST	4 (6.1)	22 (33.3)	35 (53)	5 (7.6)	66
CUS	2 (2.4)	38 (45.8)	41 (49.4)	2 (2.4)	83
CUK	5 (9.3)	10 (18.5)	35 (64.8)	4 (7.4)	54
Total	22 (7.1)	109 (34.9)	165 (52.9)	16 (5.1)	312

Values in the parentheses indicate percentage.

Table 6. University-wise knowledge of terminology used in databases

University	Google	Encyclopaedia	Dictionary	Thesaurus	Total
UoK	96 (85.7)	5 (4.5)	3 (2.7)	8 (7.1)	112
IUST	43 (65.2)	8 (12.1)	4 (6.1)	11 (16.7)	66
CUS	54 (63.5)	16 (18.8)	7 (8.2)	8 (9.4)	85
CUK	43 (79.6)	5 (9.3)	2 (3.7)	4 (7.4)	54
Total	236 (74.4)	34 (10.7)	16 (5)	31 (9.8)	317

Values in the parentheses indicate percentage.

Knowledge of library catalogue

The study reveals that more than half of the faculty members (52.9%) are aware about library catalogue being used to find a book in a library.

Batabyal (2018) found 53.8% respondents aware of library catalogue while as Lata (2014) found most of the faculty members (85.8%) aware about library catalogue.

University-wise response to the question revealed that CUK (64.8%) faculty members are much aware about the library catalogue than other three university faculty members (Table 5).

It is inferred from the results that about half of the faculty members only are aware about library catalogue.

Knowledge of terminology used in databases

The study discloses that the faculty members are mostly unaware of thesaurus being used to find information while searching a database. A meagre percentage of 9.8% faculty members have given correct answer to the question.

Lata (2014) has revealed a better situation in which 35.6% faculty members were found to be aware about thesaurus.

University-wise response to the question revealed that faculty members of IUST (16.7%) are more aware of thesaurus than other three universities (Table 6).

It is inferred that the faculty members are not aware of thesaurus.

Searching with different combination of words

The results of the study reveal that more than half of the faculty members (56.5%) have given correct answer to the question.

University-wise response to the question revealed that percentage of correct response is highest in UoK (63.4%) followed by CUS, IUST and CUK, respectively (Table 7).

Boolean operators

Most of the faculty members (59.6%) are aware of Boolean operators. However, a good percentage (29.3%) of them don't know about the concept while others gave wrong answer to the question. However, the respondents of study of Sarman (2013) are not aware of Boolean operators as only 25.9% gave correct answer to the question asked to them while as Lata (2014) found also only 30.8% faculty members aware about Boolean operators. However, Batabyal (2018) found 49.2% respondents aware of Boolean operator but Rehman and Alfaresi (2009) revealed that only 20.7% respondents have knowledge about Boolean operators. Similarly, Bhat and Ganaie (2016) conducted a study on the users of the Dr Y.S. Parmar University of Horticulture and Forestry. The study reveals that less than 50% of users are aware about Boolean operators.

University-wise response to the question shows that more faculty members of IUST (69.7%) know about the Boolean operators than other three universities under study (Table 8).

It is inferred that only about half of the faculty members are aware of Boolean operators. However, female faculty members are more aware about Boolean operators than male faculty members.

Knowledge of bibliography

The study reveals that almost half of the faculty members (49.5%) know how to find related articles. However, more than half of the faculty members are still not aware of the best way to find related articles using bibliography. However, Lata (2014) found around 69.7% faculty members aware about bibliography as a source of finding related articles while as the study of Sarman (2013) discloses that only 31.4% respondents are familiar with bibliography. Likewise, Batabyal (2018) found 35.5% respondents aware of bibliography.

On comparing the answers given by faculty members the study reveals that CUK (72.2%s) faculty members are much aware of bibliography as a source of finding related articles followed by IUST, UoK and CUS, respectively (Table 9).

It is inferred that only about half of the faculty members are aware about bibliography. However, faculty members of CUK are more aware about the same than other three universities.

Knowledge of wildcards/ truncation

The results show that only a meagre percentage of 13% faculty members know about wildcards/ truncation. Similar sort of situation also has been reported by Sarman (2013) whereunder only 4.4% of the respondents replied correctly to the question posed related to wildcards. However, Batabyal (2018) found 31.9% respondents aware of wildcard.

Table 7. University-wise knowledge of searching with different combination of words

University	COVID-19 prevention	Prevention of COVID-19	COVID-19, prevention	All of them	Total
UoK	23 (20.5)	16 (14.3)	2 (1.8)	71 (63.4)	112
IUST	16 (24.2)	14 (21.2)	1 (1.5)	35 (53)	66
CUS	13 (15.3)	22 (25.9)	3 (3.5)	47 (55.3)	85
CUK	7 (13)	18 (33.3)	3 (5.6)	26 (48.1)	54
Total	59 (18.6)	70 (22.1)	9 (2.8)	179 (56.5)	317

Values in the parentheses indicate percentage.

Table 8. University-wise knowledge of Boolean operators

University	COVID-19 OR air particle	COVID-19 AND air particle	COVID-19 NOT air particle	Don't know	Total
UoK	14 (12.5)	64 (57.1)	2 (1.8)	32 (28.6)	112
IUST	8 (12.1)	46 (69.7)	0	12 (18.2)	66
CUS	5 (5.9)	48 (56.5)	3 (3.5)	29 (34.1)	85
CUK	3 (5.6)	31 (57.4)	0	20 (37)	54
Total	30 (9.5)	189 (59.6)	5 (1.6)	93 (29.3)	317

Values in the parentheses indicate percentage.

Comparative study reveals that faculty members of IUST (24.2%) know about wildcards/ truncation more than other three universities under study (Table 10).

It is inferred from the results that the faculty members in select four universities are not aware about wildcards/ truncation.

Writing with proper citation

The response to the question reveals that most of the faculty members (79.4%) know about writing in own words so as to avoid plagiarism.

Among the four universities, CUK (85.2%) faculty members mostly answered the question correctly followed by UoK, IUST and CUS, respectively (Table 11).

It is inferred that the faculty members have knowledge about referencing and writing in own words to avoid plagiarism.

Overall scenario

The correct responses of all the ten questions were analysed to reveal the overall percentage of correct answers given by each university under study. The data reveals that the highest percentage of correct answers is given by CUK (49.44%) followed by IUST (45.61%), UoK (44.25%) and CUS (40.12%) respectively. As such the overall scenario is quite grim as the faculty members of all four universities are not able to cross the half mark of correct answers to the questions posed to them (Table 12).

Table 9. University-wise knowledge of bibliography

University	Search the database	Other issues/ volume of journal	Library catalogue	Bibliography from the article	Total
UoK	38 (33.9)	15 (134)	7 (6.3)	52 (46.4)	112
IUST	20 (30.3)	12 (18.2)	1 (1.5)	33 (50)	66
CUS	28 (32.9)	16 (18.8)	8 (9.4)	33 (38.8)	85
CUK	11 (20.4)	2 (3.7)	2 (3.7)	39 (72.2)	54
Total	97 (30.6)	45 (14.2)	18 (5.7)	157 (49.5)	317

Values in the parentheses indicate percentage.

Table 10. University-wise knowledge of wildcards/ truncation

University	Wildcard/truncation (*)	Boolean Operator (AND, OR & NOT)	Phrase searching	Don't know	Total
UoK	10 (9)	19 (17.1)	24 (21.6)	58 (52.3)	111
IUST	16 (24.2)	13 (19.7)	15 (22.7)	22 (33.3)	66
CUS	10 (11.9)	20 (23.8)	14 (16.7)	40 (47.6)	84
CUK	5 (9.3)	4 (7.4)	17 (31.5)	28 (51.9)	54
Total	41 (13)	56 (17.8)	70 (22.2)	148 (47)	315

Values in the parentheses indicate percentage.

Table 11. University-wise knowledge of writing with citation

University	Write word for word from an article without quoting author	Copy and paste with proper citation	Write in your own words quoting original author	Don't know	Total
UoK	7 (6.3)	5 (4.5)	95 (84.8)	5 (4.5)	112
IUST	3 (4.5)	10 (15.2)	52 (78.8)	1 (1.5)	66
CUS	6 (7.1)	16 (19)	58 (69)	4 (4.8)	84
CUK	3 (5.6)	3 (5.6)	46 (85.2)	2 (3.7)	54
Total	19 (6)	34 (10.8)	251 (79.4)	12 (3.8)	316

Values in the parentheses indicate percentage.

Table 12. Overall percentage of correct answers

University	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Total	Percentage
UoK	46	80	20	54	8	71	64	52	10	95	500	44.25
IUST	23	43	7	35	11	35	46	33	16	52	301	45.61
CUS	33	55	8	41	8	47	48	33	10	58	341	40.12
CUK	22	30	29	35	4	26	31	39	5	46	267	49.44

It is, thus, inferred that most of the faculty members of all four universities require guidance regarding searching and retrieval of information either physically or online.

Relationship between universities and capability to answer correctly

In order to find out whether there is any association between the correct answers given by the faculty members of four universities, the data regarding correct answers was recoded in SPSS so as to make three groups mentioned in Table 13. The recoding was required so as to reveal the correct association between the two variables.

The chi-square test conducted on the recoded data reveals that there is insignificant association at 5% significance level between universities and correct answers given by the faculty members, $\chi^2(6, N = 318) = 8.896$, $p = .180$.

As such, it can be inferred that there is no relationship between correct answers given by the faculty members of four universities. All faculty members have responded in a similar manner in all the four universities.

User awareness programmes and capability of answering correctly

In order to find out whether there is any association between the correct answers and user awareness programmes attended by the faculty members, the data was recoded to make three groups.

The chi-square test results reveal that there is statistically significant association between the attendance in user awareness programmes and correct answers given by the faculty members. Faculty members are more likely to give correct answer if they attend such programmes. Furthermore, faculty members are more likely retrieve exact information when they attend orientation programmes. However, the Cramer's V $(0.25 > V < 0.75)$ indicates that there is moderate association between the two variables, $\chi^2(2, N = 318) = 88.312$, $p = .000$ (Table 14).

Table 13. University-wise relationship of answers

	Number of correct answers				
University	1–3	4–6	7–10	Total	$\chi^2(6)$
UK	33	66	14	113	8.896*
IUST	19	37	10	66	
CUS	35	43	7	85	
CUK	12	31	11	54	
Total	99	177	42	318	

*$p > .05$.

Table 14. Relationship between awareness programmes and correct answers

	Number of correct answers				
Attended any user awareness programme of library	1–3	4–6	7–10	Total	$\chi^2(2)$
No	58	19	2	79	88.312*
Yes	41	158	40	239	
Total	99	177	42	318	

*$p < .05$, Cramer's $V = .527$.

It can easily be inferred that if faculty members are provided proper orientation, they can search the required information precisely resulting in saving their precious time and other resources.

Discussion

Since the advent of Internet, there has been a tremendous growth of information. This has created problems to the scholarly community as they are unable to find a specific information in case, they are information illiterate. Furthermore, it is hard for the information illiterate people to trust the abundant sources of information available to them. In this scenario Mackey and Jacobson (2004) have rightly suggested that information literacy is essential in the modern

era of information age. Likewise, Boeriswati (2012) argued that people can get the authentic information only when they know how to search.

The data related to basic information sources reveals that only 39.4% faculty members in Jammu & Kashmir are aware about encyclopaedias providing basic information related to a concept but majority of them are unaware of encyclopaedias. Contrary to this, Batabyal (2018) found 64% respondents aware of finding background information from encyclopaedias in West Bengal but Lata (2014); Rehman and Alfaresi (2009) found 49.8% and 33.7% population respectively aware of encyclopaedias and thus confirming the results of this study.

Usually, faculty members are well aware of journals providing nascent information in any field of study as the same source has significant positive effect on research productivity (Ani et al., 2015). However, the results indicate that only 65.8% of the faculty members are aware about journals providing current information. Similar results were also found in the study of Sarman (2013) who found only 56.3% of the respondents aware about journals providing current information while as a higher percentage of 77.3% respondents in the medical colleges of Haryana, Punjab and Chandigarh are capable of searching current information in journals (Lata, 2014).

Users in a library try to search books through various ways. However, the technical aspects surely help in retrieving the books quickly and thereby saving the time of the users. The study reveals that although about half of the faculty members are aware about library catalogue similar to the study of Batabyal (2018) and only 20.2% are aware about call number. Contrary to this, Lata (2014) found most of the faculty members (85.8%) aware about library catalogue but similar sort of situation is seen with call number which is known to only 28.9% of respondents.

For higher studies and research, numerous databases are available and it is often recommended to search these databases with particular thesaurus. Keeping this thing in mind, the faculty members were posed a simple question to reveal their knowledge regarding thesaurus. However, the study discloses that the faculty members are mostly unaware of thesaurus. A meagre percentage of faculty members (9.8%) know about thesaurus. Contrary to this finding, Lata (2014) revealed a better situation in which 35.6% faculty members were found to be aware about thesaurus in medical colleges of Haryana, Punjab and Chandigarh. Likewise, a meagre percentage of faculty members (13%) know about wildcards/ truncation. Similar sort of situation also has been reported by Sarman (2013) while as Batabyal (2018) found 31.9% respondents aware of wildcards.

While searching for information, people often use Boolean Operators, search with different combination of words and make use of bibliography to find relevant literature. Regarding all these questions, around half of the faculty members are found to aware of them similar to the findings of Bhat and Ganaie (2016); Batabyal (2018) regarding use of Boolean Operators. However, the study of Sarman (2013); Lata (2014); Rehman and Alfaresi (2009) reveal that the respondents are mostly unaware of Boolean operators. Regarding bibliography, Lata (2014) found higher percentage of faculty members (69.7%) aware about it while as Sarman (2013); Batabyal (2018) carried out studies in which respondents are mostly seen unaware of bibliography.

Although with the advent of Internet, research has seen a tremendous increase but at the same time some issues also came into existence in the form of plagiarism. A question was posed to the faculty members to see whether they are aware about this issue or not. The response to the question reveals that most of the faculty members (79.4%) know how to avoid plagiarism.

On the one hand, the libraries need to enhance their collection, services, infrastructure and the atmosphere of the libraries while as on the other hand information literacy programmes are equally important for the faculty members in all universities. The results suggest that continuous orientation programmes should be the focus of libraries so that the faculty members are benefitted since there is statistically significant association between the user awareness programmes and correct answers given by the faculty members. This will ultimately result in better research output of the universities and higher ranking as well.

Conclusion

The study concludes with remarks that it is essential to have information literacy (IL) because of abundance of information available (Atikuzzaman and Ahmed, 2023). Information literacy forms a key concept not only within Library and Information Science (Hicks et al., 2023) but in Computer Education as well. By utilising IL skills, users would not have to depend on others to solve their problems; instead, they learn in a genuine and autonomous manner. Greater responsibility for their own education is fostered by this, and

it will help them become more independent, imaginative, critical thinkers, and learners. The study aimed to answer the simple question whether faculty members in selected universities of J&K know how to search or not. The results make it clear that the faculty members are not fully aware about various information searching techniques. Although some faculty members know how to search for a document in a library but others are unaware about catalogue and call number. If the universities in J&K have to make an impact at the national and international level, they have to take urgent steps for the information literacy skills of faculty members which will eventually led to higher research yield. The better research output will ultimately give recognition to the universities globally. These skills must include computer and Internet literacy skills. The efforts are to be taken by the libraries since librarians have a long history of involvement with teaching research skills to library users (Pullman, 2016). The support is also required from the faculty members since cooperation in this regard is must. The faculty members have to demonstrate broad-mindedness in this regard so that they are benefitted which will eventually benefit the scholars and students of their universities as well.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

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